

BREAK THE LIMIT

Agenda

- ① Columnar Storage
- ② Partitions
- ③ SQL Interviews

Columnar Storage employees

id	first_name	last_name	customer_zip
1	Jane	Connor	22801
2	Mmanuel	Diaz	22821
3	Bob	Wilson	22821
4	Deanna	Washington	22801
5	Abigail	Harris	22801

Traditional Row Based Storage

Row 1	1	Jane	Connor	22801
Row 2	2	Mannuel	Diaz	22821
Row 3	3	Bob	Wilson	22821
Row 4	4	Deanna	Washington	22801
Row 5	5	Abigail	Harris	22801

are all datatypes
same in a single
row

No

Columnar Storage

id	1	2	3	4	5
first_name	Jane	Mannuel	Bob	Deanna	Abigail
last_name	Connor	Diaz	Wilson	Washington	Harris
last_zip	22801	22821	22821	22801	22801

Comparison

Feature	Row - Based	Columnar
Use Case	OLTP MySQL	OLAP Big Query
Insert / update	Faster	Slower
Aggregated	Slower	Faster
Compression	negligible	v. high

Run length encoding , Dict encoding , Delta encoding

cust - zip

22801	22821	22821	22801	22801
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Run length encoding

22801 , 22821 x 2 , 22801 x 2

Dict encoding

2

1 : 22801

customer zip

2 : 22821

1 , 2 , 2 , 1 , 1

3

Delta encoding

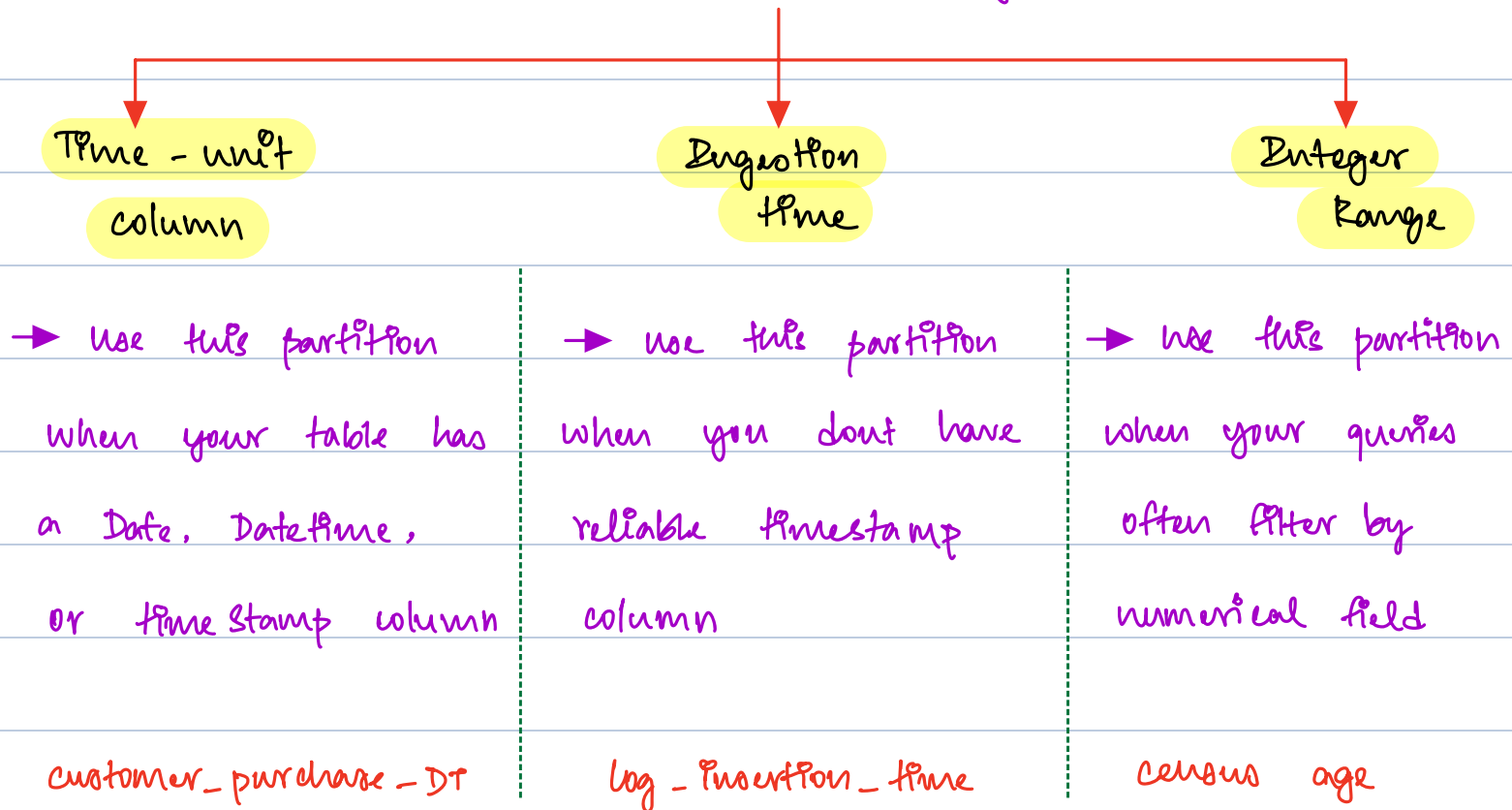
id 1 , 2 , 3 , 4 , 5

id 1 , +1 , +1 , +1 , +1

Partitions - Big Query

- ① Partitions is a database process to divide very large tables into multiple smaller, individual parts
- ② Scan lesser data
- ③ Reduces query execution time

Types of Partitions in Big Query



Break 10:05 → 10:12 pm

SQL Interviews

- farmers market Dataset
- share a question / tables necessary
- write the query
 - no of rows
 - first column value

cust Id	market	cost	prodn
1	25th	25	1
2	26th	100	2
1	25th	100	2
2	27th	75	3
1	26th	75	3

Saturday Sunday — solve all these problems

Sunday — solution in the whats App group

Partition and sharding

can be present in  100% horizontal scaling
single machine